

Experiment no. 05

Aim: Write a program to implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript. Consider the end user as one of the parties (Alice) and the JavaScript application as other party (bob).

Theory:

Diffie-Hellman algorithm:

The Diffie-Hellman algorithm is being used to establish a shared secret that can be used for secret communications while exchanging data over a public network using the elliptic curve to generate points and get the secret key using the parameters.

- For the sake of simplicity and practical implementation of the algorithm, we will consider only 4 variables, one prime P and G (a primitive root of P) and two private values a and b .
- P and G are both publicly available numbers. Users (say Alice and Bob) pick private values a and b and they generate a key and exchange it publicly. The opposite person receives the key and that generates a secret key, after which they have the same secret key to encrypt.

Step-by-Step explanation is as follows:

Alice	Bob
Public Keys available = P, G	Public Keys available = P, G
Private Key Selected = a	Private Key Selected = b
Key generated =	Key generated =
Exchange of generated keys takes place	
Key received = y	key received = x
Generated Secret Key =	Generated Secret Key =
Algebraically, it can be shown that	

Alice	Bob
Users now have a symmetric secret key to encrypt	

Example:

Step 1: Alice and Bob get public numbers $P = 23$, $G = 9$

Step 2: Alice selected a private key $a = 4$ and

Bob selected a private key $b = 3$

Step 3: Alice and Bob compute public values

Alice: $x = (9^4 \bmod 23) = (6561 \bmod 23) = 6$

Bob: $y = (9^3 \bmod 23) = (729 \bmod 23) = 16$

Step 4: Alice and Bob exchange public numbers

Step 5: Alice receives public key $y = 16$ and

Bob receives public key $x = 6$

Step 6: Alice and Bob compute symmetric keys

Alice: $k_a = y^a \bmod p = 65536 \bmod 23 = 9$

Bob: $k_b = x^b \bmod p = 216 \bmod 23 = 9$

Step 7: 9 is the shared secret.

Conclusion:

Thus we have studied Diffie-Hellman Key Exchange mechanism using HTML and JavaScript